LAHORE AIR QUALITY INDEX (AQI) FORECASTING SYSTEM

A Machine Learning-Based Three-Day AQI Prediction System

Project Report

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1. Introduction

Air pollution is a major environmental and health problem, especially in large cities like Lahore. Air quality can change significantly due to pollution sources, weather conditions, and seasonal changes. Therefore, predicting air quality in advance can help people take necessary precautions.

This project develops a machine learning-based system to predict Lahore's Air Quality Index (AQI) for the next three days. The system collects weather and air pollution data from Open-Meteo, processes the data, and calculates the AQI using the US EPA method. Machine learning models then use current and historical data to forecast future AQI levels.

The complete system is automated using GitHub Actions, while Feast is used for feature management and Streamlit provides a web dashboard. The dashboard displays the current AQI, three-day forecast, pollution levels, historical trends, and health-related alerts.

2. Objectives

The main objectives of this project are:

- To predict Lahore's AQI for the next 1, 2, and 3 days.
- To collect and process air quality and weather data automatically.
- To use machine learning to improve AQI forecasting.
- To automate model training and data updates using GitHub Actions.
- To provide AQI forecasts and health alerts through a Streamlit dashboard.
- To identify the important factors that affect AQI predictions.

3. Data

The project uses air quality and weather data from Open-Meteo for Lahore. The dataset contains hourly records of major pollutants, including PM2.5, PM10, CO, SO_2, NO_2, and O_3. It also includes weather information such as temperature, humidity, pressure, wind speed, wind direction, and cloud cover.

The data covers approximately August 2022 to August 2026, providing more than 35,000 hourly records. These records are aggregated into daily data for machine learning. The US EPA AQI formula is used to calculate the daily AQI from pollutant concentrations.

The processed data is then used to create additional features such as previous AQI values, PM2.5 history, rolling averages, and seasonal information. These features help the machine learning models identify patterns and predict AQI for the next three days.

4. Data Pipeline

The project uses an automated pipeline to collect, process, and prepare data for AQI prediction. First, **Open-Meteo** provides hourly air quality and weather data. The data is then processed and converted into daily records, followed by AQI calculation and data cleaning.

Next, **feature engineering** creates useful features such as AQI and PM2.5 lag values, rolling averages, and seasonal information. These features are stored using **Feast** and used for training the machine learning models.

The pipeline is automated using **GitHub Actions**. The data collection and feature update process runs hourly, while the machine learning models are retrained daily using the latest available data. The updated models are then used by the Streamlit dashboard to provide the latest AQI forecasts.

5. Feature Engineering & Data Cleaning

First, the raw hourly data is converted into daily data. The average value of each pollutant and weather measurement is calculated for every day. The daily AQI is then recalculated from the daily average pollutant concentrations using the US EPA AQI method.
The data is cleaned by filling missing numerical values using the previous available value and removing invalid AQI values outside the 0–500 range.

After cleaning, new features are created to help the model understand pollution patterns. These include previous AQI and PM2.5 values from 1, 2, 3, and 7 days ago, rolling averages over 3, 7, and 14 days, seasonal features using sine and cosine, weather rolling averages, and the daily change in AQI.
Finally, three target columns are created: tomorrow's AQI, AQI after two days, and AQI after three days.

The cleaning stage then keeps only rows where all required features and future target values are available for training. The latest complete feature row is separately saved for making the current three-day forecast.

6. Feature Store

The project uses Feast as a feature store to manage the data used by the machine learning models. After feature engineering, the processed daily data is saved as a Parquet file inside the feature_repo/data folder. A city_id field is also added to identify the data as belonging to Lahore.

The project then runs feast apply, which registers the feature definitions in the Feast repository. This allows the engineered features to be organized and made available for the machine learning pipeline.

In simple terms, Feast acts as a central place for storing and managing the features that are prepared for model training and prediction.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the AQI patterns in Lahore before training the machine learning models. The analysis focuses on the AQI trend over time, monthly seasonality, day-of-week patterns, relationships between AQI and other variables, and the distribution of AQI values.

7.1 AQI Trend Over Time

The AQI trend plot shows how air quality changes throughout the dataset period. It helps identify periods of high and low pollution and shows the overall variation in Lahore's AQI over time.

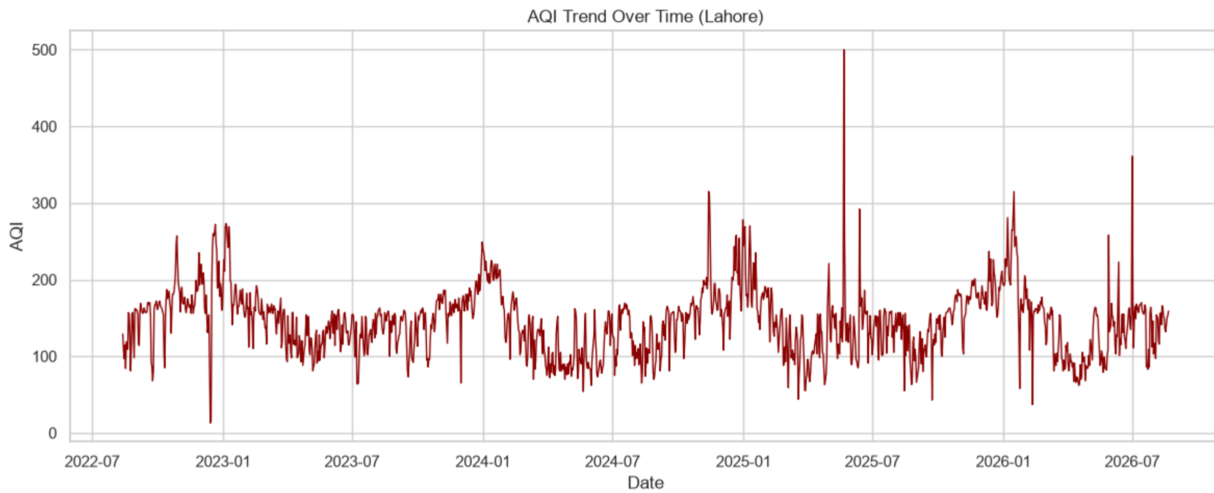

Figure 1 AQI Trend Over Time

7.2 AQI by Month

A monthly analysis was performed to check seasonal patterns. The AQI values are compared across all 12 months using box plots. This helps identify which months generally have higher or lower pollution levels.

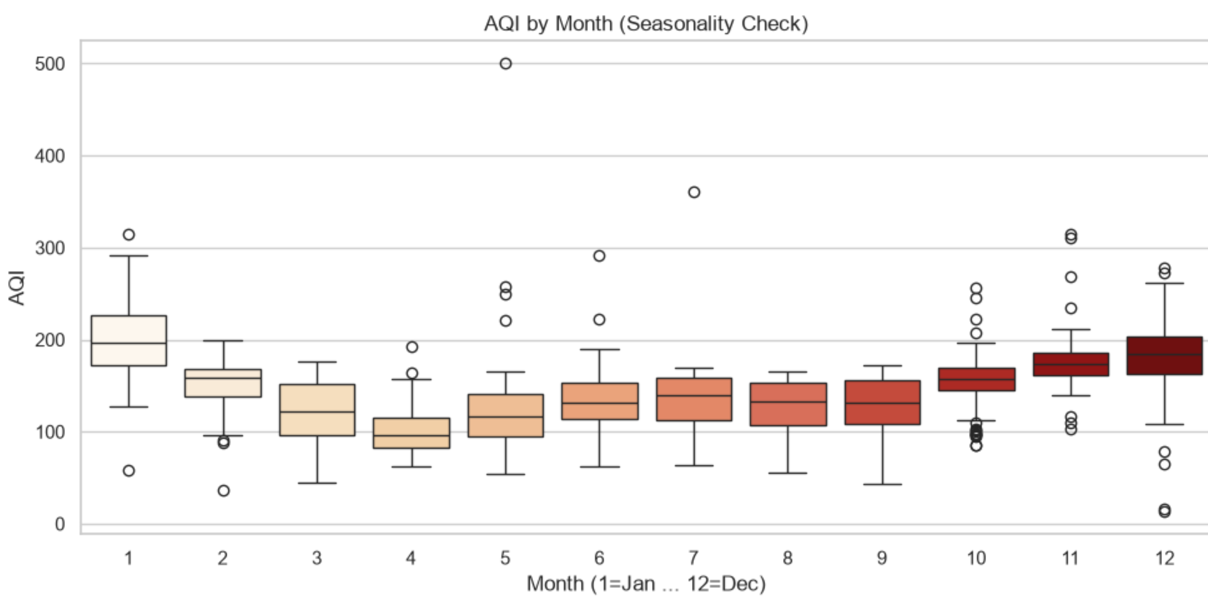

Figure 2 AQI by Month

7.3 AQI by Day of Week

AQI was also compared across the seven days of the week. This analysis checks whether there are noticeable differences in pollution between weekdays and weekends.

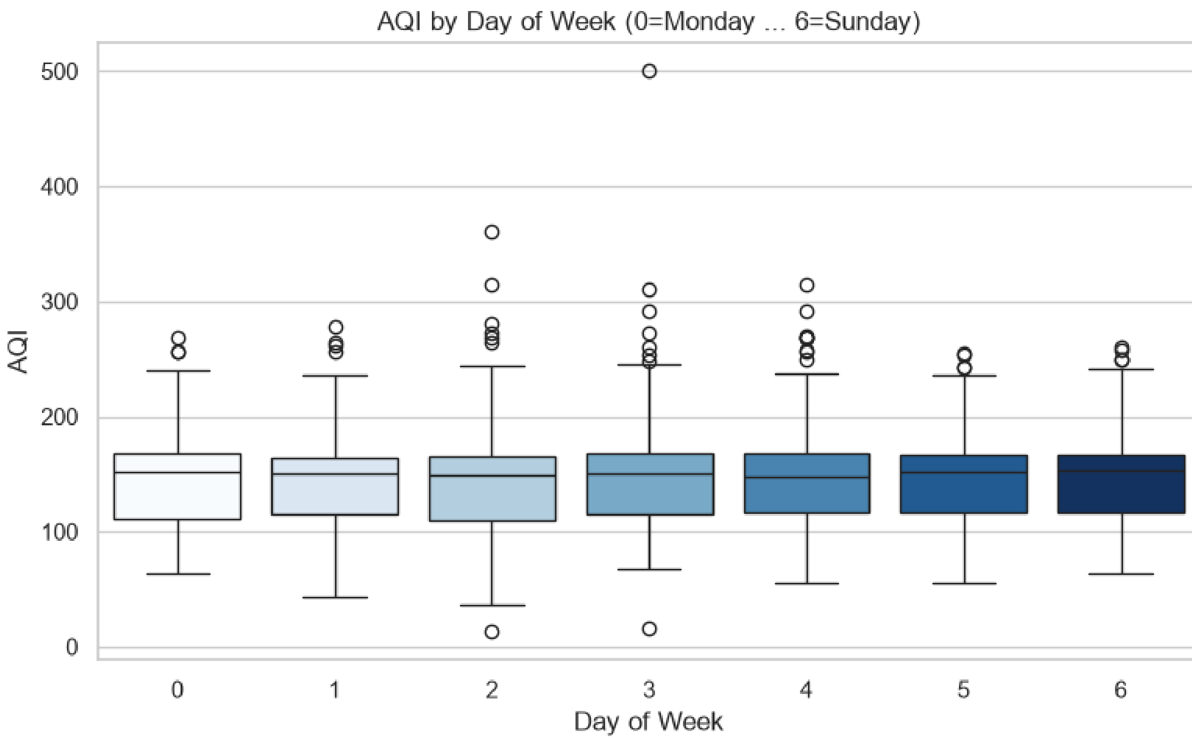

Figure 3 AQI by Day of Week

7.4 Correlation Analysis

A correlation heatmap was created to examine the relationship between AQI and important pollutant and weather variables. The analysis includes PM2.5, PM10, CO, SO_2, NO_2, O_3, temperature, humidity, pressure, wind speed, and cloud cover.

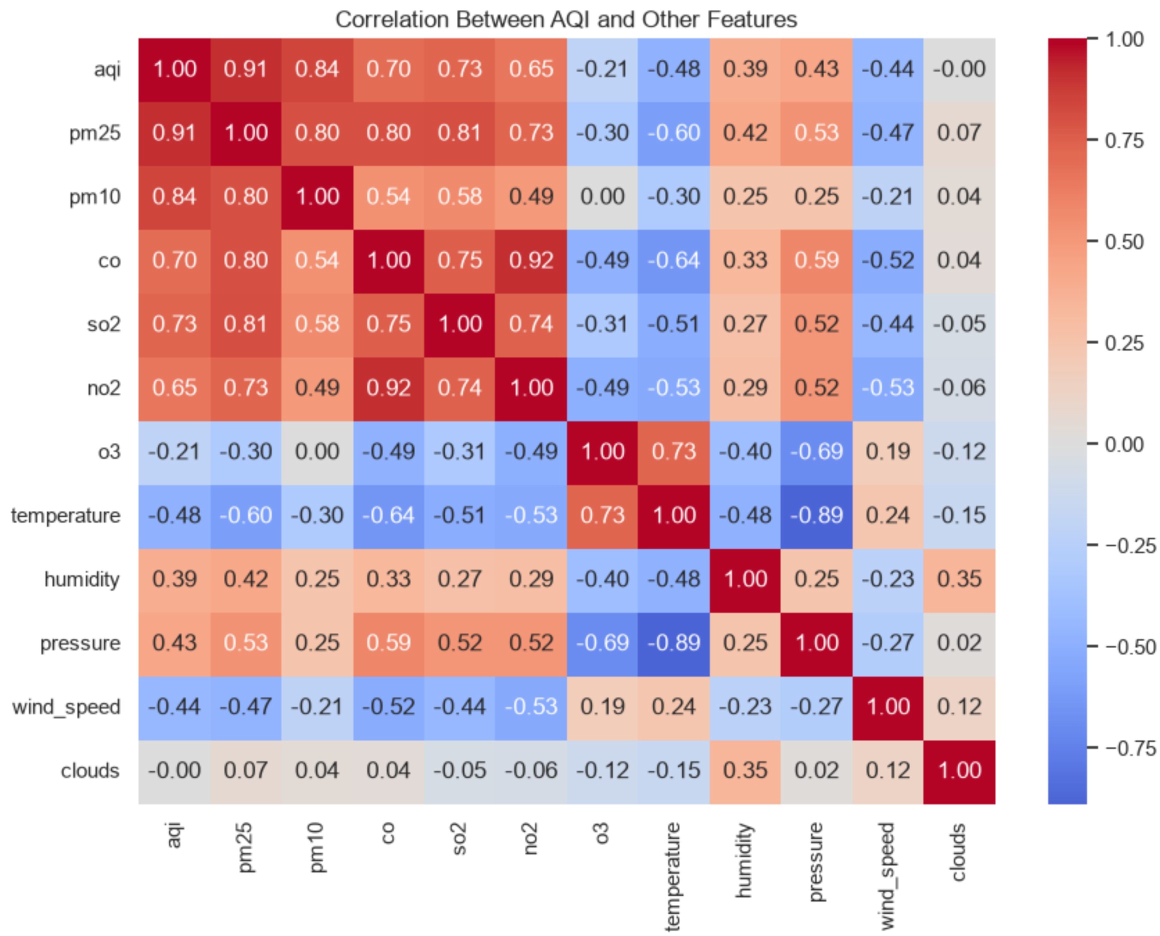

Figure 4 Correlation Analysis

7.5 AQI Distribution

A histogram was used to understand how AQI values are distributed across the dataset. It shows how frequently different AQI levels occur and helps identify the typical range and extreme pollution values.

Overall, the EDA helps identify important patterns in the data and provides a better understanding of the factors that can be useful for AQI forecasting.

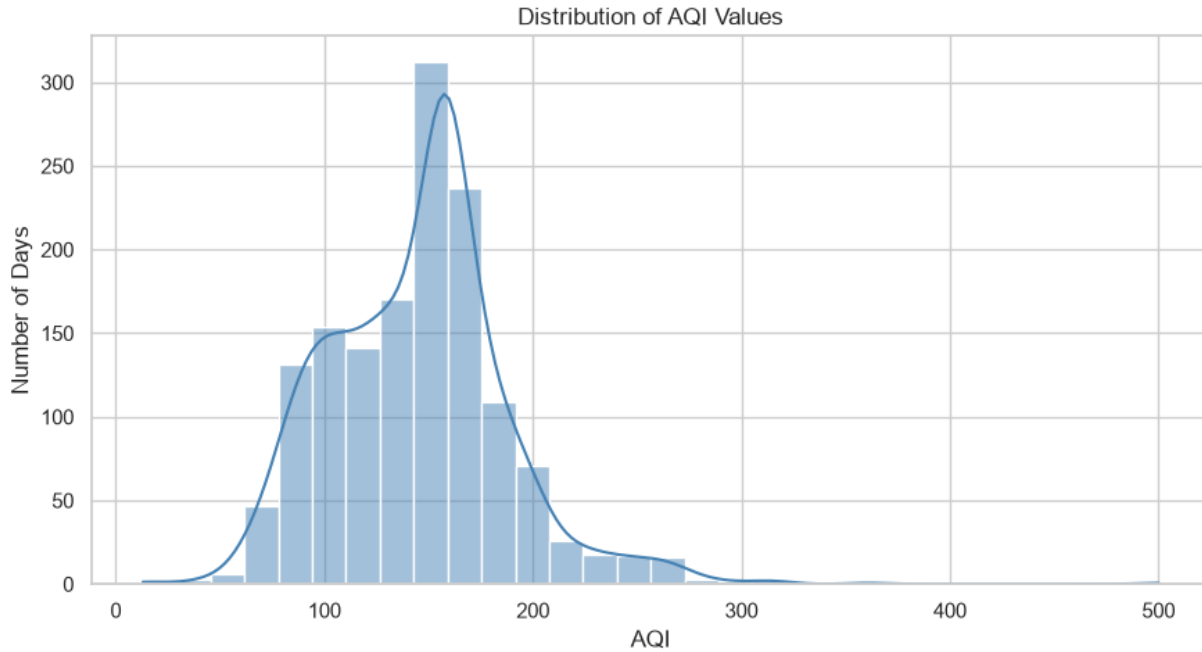

Figure 5 AQI Distribution

8. Modeling Approach

The project uses a separate machine learning model for each forecasting horizon. Three models are trained to predict the AQI for 1 day, 2 days, and 3 days ahead.

Two main algorithms, Random Forest and XGBoost, are tested. Their parameters are tuned using Randomized Search with Time Series Cross-Validation. Since the data is time-based, the model is not evaluated using random shuffling. Instead, the data is divided according to time so that future information does not leak into the training process.

The most recent 20% of the data is kept as a final test set. The model that performs better during validation is selected for each forecasting horizon. The final models are then trained using the available training data and saved for use by the dashboard.

This approach allows the system to make separate and reliable predictions for tomorrow, two days ahead, and three days ahead.

9. Model Training & Selection

The cleaned data is retrieved from the Feast feature store and divided into training and testing sets. The most recent 20% of the data is kept for testing, while the earlier 80% is used for training. This keeps the evaluation in chronological order and prevents future information from entering the training-data.

For each forecasting target (Day +1, Day +2, and Day +3), the system tests Random Forest and XGBoost models. Randomized Search is used to try different model parameters, while 5-fold Time Series Cross-Validation is used to compare their performance using R^2.

The model with the best cross-validation R^2 score is selected separately for each forecasting horizon. The selected models are then evaluated on the unseen test data using MAE, RMSE, and R^2.

Finally, the trained models, evaluation results, selected parameters, and feature information are saved as model artifacts. SHAP is also used to identify which features have the greatest influence on each prediction.

10. Results & Evaluation

The trained models were evaluated on the most recent 20% of the dataset, which was kept completely separate from training. Three metrics were used: R^2, MAE, and RMSE.

Forecast	**R^2**	**MAE**	**RMSE**
Day +1	0.671	17.61	26.94
Day +2	0.541	22.25	31.84
Day +3	0.506	23.94	33.03

The results show that the model performs best when predicting one day ahead, with an R^2 of 0.671. Performance decreases for the second and third days, which is expected because longer-term predictions are more difficult.

The MAE values show that the typical prediction error is about 17 AQI points for Day +1, increasing to about 24 AQI points for Day +3. Overall, the results show that the system can provide useful three-day AQI forecasts while maintaining a time-based and realistic evaluation method.

11. Model Improvement Experiments

Several experiments were performed to check whether the forecasting model could be improved. Different model types, target formulations, ensembles, and additional feature-engineering approaches were tested using the same chronological test setup.

Different models such as XGBoost, Extra Trees, HistGradientBoosting, and Ridge were tested. Delta modeling, log transformation, and model ensembles were also investigated. These approaches produced only small changes and did not provide a reliable improvement over the existing Random Forest model.

Additional features were also tested, including PM10 history, weather-history features, volatility/momentum features, long-term features, and feature interactions. The best feature combination gave only a small improvement on the recent test set, while removing features reduced performance.

To make the final decision more reliable, the promising approaches were tested using an expanding-window walk-forward backtest. Although some approaches performed slightly better on the recent test set, the differences were not consistent across multiple time periods. Therefore, the experiments did not provide enough evidence to replace the current Random Forest model.

The final conclusion was to keep the existing model. The experiments suggest that with the current amount of data, adding more features or changing the model provides only limited benefits. Collecting more historical data, especially more winter pollution data, is considered a more useful way to improve future performance.

12. Dashboard

The project includes a Streamlit dashboard that provides a simple interface for viewing the air quality information and model predictions. It displays the current AQI, AQI category, pollutant levels, weather conditions, and a three-day AQI forecast.

The dashboard also shows historical AQI trends and the model's important features using SHAP, helping users understand the factors affecting the predictions. Health-related warnings are displayed based on the current AQI level.

Overall, the dashboard makes the machine learning results easier to understand and allows users to view the current air quality and future AQI predictions in one place.

13. Deployment & Automation

The system is designed as an automated pipeline using GitHub Actions. New air quality and weather data is collected automatically every hour and processed through the data and feature-engineering pipeline.

The machine learning models are retrained daily using the latest available data. The updated models and data are then made available to the Streamlit dashboard. This automation reduces manual work and keeps the forecasting system updated with recent information.

14. Limitations & Future Work

The main limitation of the project is the limited amount of historical data, which can restrict the ability of machine learning models to learn complex long-term pollution patterns. Forecast accuracy also decreases as the prediction period increases from one day to three days.

Future work can focus on collecting more historical data, especially winter and smog-season data, to improve the models. More advanced forecasting techniques and additional environmental factors can also be explored to further improve prediction accuracy.

15. Conclusion

This project developed an automated three-day AQI forecasting system for Lahore using air quality data, weather information, and machine learning. The system includes data collection, AQI calculation, feature engineering, model training, evaluation, and a Streamlit dashboard.

The final Random Forest models provide useful forecasts for one, two, and three days ahead. The experiments also showed that alternative models and additional features did not provide a reliable improvement. Overall, the project demonstrates how machine learning can be combined with an automated data pipeline to provide practical air quality forecasts and health-related information.